

Day 2 — Prediction Models: Match Winner & Top Player Models

Project Title

AFL Prediction System — Match Winner and Top Player Prediction Models

Scenario

The objective of this project is to develop the core machine learning prediction models for an Australian Football League (AFL) prediction system. The previous stage of the project focused on understanding the available AFL datasets, checking data quality, defining prediction targets, and constructing meaningful historical features. After completing those preparation steps, the next stage focuses on transforming the prepared data into practical predictive models.

The scenario for this project is based on a future AI-powered AFL prediction assistant. The assistant should eventually be able to answer questions related to upcoming AFL matches, such as which team is more likely to win a particular match and which player is expected to perform strongly. To support this functionality, two different prediction systems are required. The first system predicts the outcome of an AFL match, while the second system predicts player performance and uses those predictions to identify the most likely top-performing player.

The match-winner prediction problem is treated as a classification problem because the objective is to determine the outcome of a match. The top-player problem is treated as a regression and ranking problem because player performance is represented numerically through statistics such as disposals. The regression model predicts the expected disposal count of individual players, and the resulting predictions are then used to rank players within a match.

The models developed in this stage are intended to become reusable components of a larger AI agent. Therefore, the project does not only focus on achieving predictive performance. It also focuses on preventing data leakage, evaluating model reliability, understanding feature importance, performing sanity checks, and packaging the final models into reusable prediction functions.

The overall workflow begins with simple baseline models. These baselines establish a minimum performance level that the machine learning models should attempt to improve upon. More advanced models are then trained and evaluated using chronological hold-out data. Feature importance and sanity checks are performed after evaluation to determine whether the models are using sensible historical information. Finally, the trained models are saved as reusable artifacts and prediction functions are prepared for future integration into an API or AI chat agent.

Task 1 — Baseline Models

Task Objective

The first task was to establish baseline prediction models for both prediction problems. A baseline is a simple prediction strategy that does not require a sophisticated machine learning algorithm. Its purpose is to provide a reference point against which the performance of the actual machine learning models can be compared.

Without a baseline, it would be difficult to determine whether a trained machine learning model is genuinely useful. A model could achieve a seemingly reasonable accuracy simply because the dataset contains an imbalanced target. Therefore, the baseline establishes the minimum level of performance that the later models should attempt to exceed.

For the match-winner problem, a majority-class strategy was used. The baseline predicts the most common match outcome in the available training data for every hold-out match. This provides a simple benchmark for match-result classification.

For the top-player problem, a historical leader strategy was used. The baseline uses the historical season-average or previously identified leader as a simple prediction of the player who may perform strongly in future matches. This approach does not dynamically model individual player performance and therefore provides a deliberately simple reference point.

Match Winner Baseline

The match-winner baseline was designed around the match outcome target created during the previous stage. The match result contains the categories `home_win`, `away_win`, and `draw`.

The baseline predicts the most frequently occurring class from the training data. This means that instead of analyzing team strength, recent form, opponent history, or home-ground advantage, the baseline simply assumes that the most common historical outcome will continue to occur.

The baseline was evaluated on the hold-out data using accuracy, weighted F1 score, and multiclass ROC AUC. These metrics provide different perspectives on classification performance.

Accuracy measures the proportion of matches for which the baseline predicted the correct outcome. Weighted F1 considers both precision and recall while accounting for the frequency of each class. ROC AUC evaluates how well the predicted class probabilities distinguish between the possible outcome classes.

The baseline results establish the minimum performance level that the Logistic Regression and Gradient Boosting models must be compared against.

Top Player Baseline

The top-player baseline was designed using historical player performance. The objective was to provide a simple prediction of which player might finish as the top disposal-getter.

The baseline does not train a machine learning model to understand the relationship between previous performance and future performance. Instead, it uses a simple historical leader strategy. The resulting predicted player is then compared with the actual top disposal-getter in the hold-out matches.

Because the practical objective is to identify a top player rather than simply predict a numerical statistic, Top-1 and Top-5 hit rates are particularly useful evaluation measures.

Top-1 hit rate measures whether the baseline correctly identifies the actual highest-disposal player. Top-5 hit rate measures whether the actual top player appears among the five players predicted by the baseline to have the highest performance.

This baseline provides the reference point for evaluating the Random Forest regression model developed later.

Conclusion

Task 1 established simple and transparent benchmark models for both prediction problems. These baselines provide a meaningful reference against which the machine learning models can be evaluated.

The match-winner baseline establishes the performance level obtained by repeatedly predicting the most common match outcome. The top-player baseline establishes the performance level obtained using a simple historical player-leader strategy.

These baselines are important because the objective of the following tasks is not simply to train a complex model, but to determine whether machine learning provides additional predictive value over simple historical assumptions.

Task 2 — Build Match Winner Model

Task Objective

Task 2 focused on developing a machine learning model capable of predicting AFL match outcomes using historical and contextual features.

The match-winner problem was formulated as a multiclass classification problem with three possible outcomes: `home_win`, `away_win`, and `draw`.

The model development process was designed to prevent information from the future from entering the training features. Final scores, margins, and final match results were excluded because these variables are only known after the match has been completed. Including them would allow the model to directly access information that it is supposed to predict.

A chronological training and hold-out strategy was used. Earlier matches were used for model training, while later matches were reserved for evaluation. This is more appropriate for a prediction problem because the model is expected to predict future matches rather than randomly selected historical matches.

Feature Preparation

The match prediction model was constructed using the features developed during the previous stage of the project.

The feature set included information related to teams, opponents, match context, recent performance, previous results, scoring behaviour, margins, and other historical characteristics that could reasonably be available before a match takes place.

Direct outcome variables such as final score, opponent score, margin, and match result were excluded from the predictive features. This was an important step because including these variables would introduce target leakage.

The dataset was separated chronologically into training and hold-out portions. The training portion was used to fit the models, while the hold-out portion was kept unseen until the final evaluation.

Preprocessing Pipeline

An sklearn preprocessing pipeline was implemented so that numerical and categorical variables could be handled correctly.

Numerical features were processed using median imputation followed by standard scaling. Median imputation was used to handle missing numerical values without allowing missing observations to stop model training.

Categorical features were processed using most-frequent imputation followed by one-hot encoding. The `handle_unknown="ignore"` option was used so that previously unseen categorical values would not cause prediction failure during evaluation.

Using a pipeline ensured that preprocessing was consistently applied during both training and prediction.

Logistic Regression Model

The first machine learning model was Logistic Regression.

Logistic Regression was selected because it provides a relatively interpretable classification model. Its coefficients can be examined to understand how individual features contribute to the predicted classes.

The model was configured with a higher iteration limit to allow sufficient optimization and class balancing was enabled to reduce the effect of potential class imbalance.

The model produced both class predictions and class probabilities. The probabilities were particularly useful for evaluating ROC AUC and Brier score.

Gradient Boosting Model

The second model was Gradient Boosting Classification.

Gradient Boosting was selected because it can capture nonlinear relationships and interactions between features that may not be represented adequately by a simple linear classification model.

The model was configured with multiple estimators, a controlled learning rate, and a limited tree depth. These settings provide a balance between learning useful patterns and controlling unnecessary model complexity.

Gradient Boosting was evaluated using the same hold-out dataset as Logistic Regression so that the two models could be compared fairly.

Model Evaluation

Both models were evaluated using accuracy, weighted F1 score, multiclass ROC AUC, and Brier score.

Accuracy measures the percentage of correct match-outcome predictions. Weighted F1 provides a balanced evaluation of precision and recall while accounting for class frequency.

ROC AUC evaluates the quality of the model's probability-based discrimination between the outcome classes. A higher ROC AUC indicates stronger separation between the classes.

The Brier score was used to evaluate the quality of the predicted probabilities. Unlike accuracy, which only checks whether the final class is correct, the Brier score considers how close the predicted probabilities are to the actual outcome. Lower Brier score values indicate better probability calibration.

The final comparison was made between the majority-class baseline, Logistic Regression, and Gradient Boosting.

Handling the Missing-Value Error

During model development, the match prediction pipeline initially produced an error because some input features contained missing values. Logistic Regression and several sklearn estimators cannot directly process missing numerical values.

This problem was resolved by adding SimpleImputer to the preprocessing pipeline. Numerical variables were processed using median imputation, while categorical variables were processed using most-frequent imputation.

This change allowed the complete pipeline to handle missing values automatically without manually modifying the original dataset.

This was an important improvement because the final prediction system must be robust enough to handle incomplete real-world inputs.

Conclusion

Task 2 successfully established a complete machine learning pipeline for AFL match-outcome prediction. Logistic Regression and Gradient Boosting were trained using historical match features and evaluated on chronologically separated hold-out data.

The preprocessing pipeline handled both numerical and categorical variables and included imputation to prevent missing values from causing prediction failures. Direct post-match variables were excluded to reduce target leakage.

The evaluation framework considered not only classification accuracy but also F1 score, ROC AUC, and probability quality through the Brier score.

The model comparison provides the basis for selecting the final match-winner model for later integration into a reusable prediction function.

Task 3 — Build Top Player Model

Task Objective

Task 3 focused on predicting the player most likely to achieve the highest disposal count in an upcoming AFL match.

Instead of directly creating a multiclass classification problem in which the model would need to select one player identity from all possible players, the problem was framed as a regression task.

The regression approach predicts the expected disposal count for each player. Players can then be ranked according to their predicted disposal values. The player with the highest predicted disposal count becomes the predicted top disposal-getter.

This approach was selected because disposals are naturally numerical and because the same model can produce a ranked list of players rather than only a single prediction.

Match-Level Player Identification

The original player dataset did not contain a direct `match_id` column. Therefore, a composite match key was created using information available in the dataset.

The match key was constructed using year, match date, round, and the sorted combination of team and opponent. Sorting the team and opponent names ensures that the same match can be recognized regardless of which team's perspective is being considered.

This was important for correctly grouping all player observations belonging to the same match.

The match key allowed the model evaluation stage to determine the actual top disposal-getter for each match and compare that result with the model's predicted ranking.

Historical Player Features

The regression model used historical player-performance features rather than the current match's final statistics.

Previous disposals were calculated using a one-match lag. This means that the model receives the player's previous disposal performance rather than the disposal value from the match being predicted.

A five-match rolling average of previous disposals was also calculated to represent recent player form.

Previous goals and recent goal averages were included as additional historical performance indicators. Previous tackles and recent fantasy-point averages were also included to provide additional information about player involvement and overall performance.

Career game count was included as a player experience-related feature, while team and opponent were included as match-context features.

The resulting feature set included:

`year`, `round`, `career_game_count`, `previous_disposals`, `recent_disposals_avg`, `previous_goals`, `recent_goals_avg`, `previous_tackles`, `recent_fantasy_avg`, `team`, and `opponent`.

Leakage Prevention

Leakage prevention was particularly important for the top-player model.

The current match's disposal count cannot be used as a predictive feature because that value is exactly what the model is attempting to predict.

Therefore, lagged and rolling features were constructed using shifted historical values. The use of `.shift(1)` ensures that the current match's performance does not enter its own predictive features.

The data was also split chronologically so that earlier matches were used for training and later matches were reserved for hold-out evaluation.

This approach makes the evaluation more representative of a real prediction scenario.

Random Forest Regression

A Random Forest Regression model was selected for the top-player prediction task.

Random Forest Regression combines predictions from multiple decision trees. It can model nonlinear relationships between historical player statistics and future disposal performance.

The model was configured with multiple trees, a maximum tree depth, a minimum number of samples per leaf, and a fixed random state for reproducibility.

The model was placed inside an sklearn pipeline with numerical imputation, numerical scaling, categorical imputation, and one-hot encoding.

During implementation, an issue occurred because a supposedly numerical feature contained a text value such as EF. The numeric features were therefore explicitly converted using `pd.to_numeric(..., errors="coerce")`. Invalid text values were converted to missing values and then handled by the numerical imputer.

This ensured that the Random Forest model could be trained successfully.

Regression Evaluation

The model was evaluated using Mean Absolute Error and Root Mean Squared Error.

Mean Absolute Error measures the average absolute difference between predicted disposals and actual disposals. Lower MAE indicates that the predicted disposal values are closer to the actual values.

Root Mean Squared Error gives greater weight to larger prediction errors. Lower RMSE indicates better numerical prediction performance.

However, numerical prediction accuracy alone is not sufficient for this project because the practical objective is to identify the top player.

Therefore, ranking-based metrics were also calculated.

Top-1 and Top-5 Hit Rate

The predicted disposal values were used to rank players within each match.

The Top-1 hit rate measures how often the player ranked first by the model was actually the highest-disposal player in the match.

The Top-5 hit rate measures how often the actual top disposal-getter appeared among the five players ranked highest by the model.

Top-5 hit rate is particularly useful because predicting the exact number-one player can be difficult, while successfully identifying a small group of likely high-performing players is still practically valuable.

The Random Forest model was compared against the historical baseline using these ranking metrics.

Sniff Test

A three-match sniff test was performed on held-out matches to provide a qualitative evaluation of the top-player model.

For each selected match, players were sorted according to their predicted disposal values. The predicted Top-5 list was then compared with the actual Top-5 list based on final disposal counts.

This manual inspection is useful because numerical metrics cannot always explain why a particular prediction succeeded or failed.

When the model disagreed with the actual result, the disagreement was treated as an opportunity to investigate the difference between historical player form and actual match performance.

Conclusion

Task 3 successfully developed a regression-based top-player prediction system.

The Random Forest model predicts expected disposal performance for individual players and converts those numerical predictions into a ranked list. Historical lag and rolling features were used to reduce target leakage, and chronological splitting was used to create a realistic hold-out evaluation.

The model was evaluated using MAE and RMSE for numerical prediction and Top-1 and Top-5 hit rates for practical ranking performance.

The three-match sniff test provided an additional qualitative evaluation of the model's predictions.

This task establishes the foundation for a reusable top-player prediction function that can later be integrated into the overall AFL AI prediction system.

Task 4 — Feature Importance and Sanity Checks

Task Objective

Task 4 focused on understanding how the trained models make predictions and determining whether the models rely on sensible football-related information.

A machine learning model can produce strong numerical results while still relying on inappropriate or leaked features. Therefore, feature importance analysis and sanity checks were performed before packaging the models for future use.

Match Winner Feature Importance

For Logistic Regression, the model coefficients were extracted from the trained pipeline.

The coefficients provide information about the contribution of each transformed feature to the different match-result classes. The absolute coefficient magnitude was also used to identify features with stronger overall influence.

For Gradient Boosting, feature importance values were extracted from the trained estimator.

These analyses provide a way to understand which historical variables are being used most strongly by the match prediction models.

Top Player Feature Importance

Feature importance was also extracted from the Random Forest regression model.

The analysis examined the contribution of historical player-performance features such as previous disposals, recent disposal averages, previous goals, recent goals, previous tackles, and recent fantasy performance.

These features have a logical relationship with future player performance because they describe the player's previous activity and recent form.

Team and opponent information provide additional contextual information that may affect player performance.

Football-Sense Check

A football-sense check was performed to determine whether important model features were logically connected to AFL match and player performance.

Recent team form is a sensible signal for match prediction because a team's previous performance can provide information about its current competitive strength.

Home-ground information is also a reasonable contextual variable because teams may perform differently depending on venue and home-ground conditions.

Previous player performance is similarly relevant for top-player prediction because historical player involvement can provide information about future performance.

Recent disposals, previous disposals, tackles, goals, and fantasy-point performance therefore provide reasonable signals for a player-performance prediction system.

Leakage Check

A specific leakage inspection was performed on the match-winner features.

Variables containing terms such as score, margin, result, winner, win, and loss were inspected.

Final score, opponent score, final margin, and final result are not valid prediction features because they are only available after the match has finished.

These variables were therefore excluded from the predictive feature set.

The top-player model was also inspected for potentially suspicious features. Current-match disposals, goals, fantasy points, score, margin, and result cannot be used as predictive inputs when predicting the same match.

The model instead uses historical values such as `previous_disposals` and `recent_disposals_avg`.

Three-Match Match-Winner Inspection

Three held-out matches were selected for a manual sniff test of the match-winner model.

For each match, the predicted outcome was compared with the actual outcome. Correct predictions were recorded, while disagreements were retained for investigation.

A disagreement does not automatically mean that the model is invalid. Match outcomes can depend on information that may not be fully represented in the available historical dataset.

The sniff test therefore provides a qualitative complement to the numerical evaluation metrics.

Conclusion

Task 4 provided interpretability and quality-control checks for the prediction system.

Feature importance analysis helped identify the variables that contributed most strongly to match and player predictions. The football-sense review provided evidence that historical form, player performance, team context, and opponent information are reasonable predictive signals.

The leakage checks were particularly important because final match outcomes and current player statistics would otherwise provide unrealistic information to the models.

The three-match sniff tests also provided a practical way to inspect predictions beyond numerical metrics.

Overall, Task 4 increased confidence that the models were learning from historical and contextual information rather than directly accessing future outcomes.

Task 5 — Package Models

Task Objective

The final task focused on converting the trained notebook models into reusable prediction components.

A machine learning model that only exists inside a notebook is difficult to integrate into a larger application. Therefore, the trained models were serialized and a separate prediction module was designed.

The final project structure separates the trained model artifacts from the prediction logic.

Model Serialization

The final match-winner model was saved using the joblib library.

The top-player Random Forest regression model was also saved using joblib.

The models were placed inside a dedicated models directory.

The purpose of serialization is to allow the trained models to be loaded later without retraining them from the beginning.

The saved artifacts can therefore be used by another Python application, API, web application, or AI agent.

Match Winner Prediction Function

A reusable `predict_match_winner()` function was designed to accept team names and a prediction date.

The function validates the supplied team names and date before preparing the input for the trained model.

The prediction function returns the predicted outcome, the corresponding winner or draw result, the prediction date, and the class probabilities.

The probability information is important because an AI assistant should ideally be able to provide not only a predicted result but also an indication of the model's confidence distribution across the possible outcomes.

Top Player Prediction Function

A separate `predict_top_player()` function was designed for the player-performance model.

The function accepts player data, a team, an opponent, and the statistic type.

For the current implementation, disposals are the supported statistic.

The function generates predicted disposal values for the relevant players and sorts them in descending order.

The result is therefore a ranked list of players, with the highest predicted disposal value appearing first.

This design makes the function suitable for future integration into an AI assistant that could answer questions such as which players are expected to be among the leading disposal-getters.

Input Validation

Basic input validation was included to make the prediction functions safer.

The match prediction function checks that both team names are supplied and that the two teams are not identical. It also validates the prediction date.

The top-player function checks whether player data exists, whether a team has been provided, whether the requested statistic is supported, and whether all required prediction features are available.

These checks reduce the possibility of silent failures and make error messages easier to understand.

Conclusion

Task 5 successfully transformed the trained machine learning models into reusable prediction components.

The models were serialized using joblib and separated from the notebook experimentation environment. A reusable Python module was created to provide prediction functions for match outcomes and top-player rankings.

Input validation was also incorporated to improve reliability.

The packaged models can now serve as the machine learning layer of a larger AFL prediction application or AI agent.

7. Overall Project Workflow

The complete Day 2 workflow began with baseline models and progressed toward production-oriented prediction components.

The first stage established simple benchmark strategies. These benchmarks were necessary because they provide a reference point for determining whether machine learning models provide additional predictive value.

The second stage developed two match-winner classification models using Logistic Regression and Gradient Boosting. Both models used preprocessing pipelines capable of handling numerical and categorical features as well as missing values.

The third stage developed a Random Forest regression model for player disposal prediction. The predicted disposal values were converted into rankings to identify likely top players.

The fourth stage focused on model interpretation, feature importance, leakage detection, football-sense validation, and manual sniff tests.

The fifth and final stage packaged the trained models and prediction functions so that they can later be integrated into an API, web application, or AI agent.

This workflow creates a complete machine learning pipeline from baseline development to model packaging.

8. Data Leakage and Reliability Considerations

Data leakage was treated as an important concern throughout the project.

For match prediction, final match scores, opponent scores, margins, and final results were excluded from the predictive features because these values are only known after the match.

For player prediction, current-match disposal values were excluded from the predictive features. Instead, lagged and rolling historical statistics were used.

Chronological splitting was also used rather than relying entirely on random splitting. This ensures that earlier observations are used to train the model and later observations are used to evaluate future-like predictions.

These choices make the evaluation more realistic and reduce the risk of artificially inflated performance.

9. Limitations

The developed prediction system has several limitations.

The available dataset does not provide every piece of information that could influence an AFL match. Factors such as injuries, player selection changes, weather, venue-specific conditions, tactical changes, and late team announcements may not be completely represented in the current feature set.

The top-player prediction model focuses primarily on disposal performance. A player can have a major impact on a match through goals, tackles, clearances, marks, or other statistics without being the highest-disposal player.

The model therefore represents a specific definition of top-player performance rather than a complete measure of player quality.

The absence of a direct match identifier in the player-level dataset also required the construction of a composite match key using date, round, year, team, and opponent information.

Finally, model performance should be monitored when new seasons become available because relationships between historical features and future outcomes can change over time.

10. Future Improvements

The system can be improved by adding more detailed match and player context.

Future versions could incorporate player availability, injury information, venue, weather conditions, team ladder position, recent opponent strength, player position, home-ground history, and more advanced rolling statistics.

The top-player model could also be extended to predict multiple statistics, including goals, tackles, marks, fantasy points, and clearances.

A learning-to-rank approach could also be investigated as an alternative to the current regression-and-ranking strategy.

The prediction functions could eventually be exposed through a FastAPI backend and connected to an AI chat agent. The agent could call the match-winner and top-player tools when users ask AFL prediction questions.

Model monitoring and periodic retraining could also be introduced so that the system remains updated as new AFL seasons and match results become available.

11. Final Overall Conclusion

The Day 2 AFL Prediction Models project successfully transformed the cleaned and feature-engineered AFL data from the previous stage into practical machine learning prediction systems.

The project began by establishing baseline models for both match prediction and top-player prediction. These baselines provided simple performance benchmarks and ensured that the more complex machine learning models could be evaluated against meaningful reference strategies.

For match prediction, the problem was formulated as a multiclass classification task containing home_win, away_win, and draw outcomes. Logistic Regression and Gradient Boosting models were developed using sklearn preprocessing pipelines. Numerical variables were processed through imputation and scaling, while categorical variables were imputed and one-hot encoded.

The match prediction models were evaluated using accuracy, weighted F1 score, multiclass ROC AUC, and Brier score. These metrics provided information about classification correctness, class-balanced performance, probability discrimination, and probability quality.

For top-player prediction, the problem was formulated as a regression task. Instead of directly predicting a player identity, the model predicts the expected disposal count for each player. Players can then be ranked according to their predicted disposal values.

The Random Forest regression model used historical player performance features including previous disposals, recent disposal averages, previous goals, recent goal averages, previous tackles, recent fantasy performance, career game count, team, and opponent.

Historical lag and rolling features were used to prevent the current match's performance from entering the model inputs. This was an important leakage-prevention step.

The top-player model was evaluated using MAE and RMSE as numerical regression metrics and Top-1 and Top-5 hit rates as ranking-oriented metrics. A three-match sniff test was also performed to compare model predictions with actual player performance.

Feature importance analysis was then performed for the trained models. The analysis provided insight into the historical and contextual features that influenced the predictions. Football-sense checks were used to determine whether the important features were logically related to match and player performance.

The leakage inspection confirmed that post-match information such as final scores, final margins, final results, and current-match player statistics should not be used as predictive inputs.

The final task converted the trained models into reusable artifacts. The match-winner and top-player models were serialized using joblib and a separate predict.py module was created to provide reusable prediction functions.

The resulting architecture separates model training, model artifacts, and prediction logic. This makes the system easier to maintain and provides a foundation for future deployment.

Overall, the Day 2 project establishes the core machine learning layer of the planned AFL AI prediction system. The project now contains baseline models, trained predictive models, evaluation procedures, feature-importance analysis, leakage checks, manual sanity testing, and reusable model artifacts. These components can serve as the foundation for the next stage, where the prediction models can be connected to an API and eventually exposed as callable tools for an